



Speech Anatomy 1

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Seeing Speech

Maybe you think speech looks like this:

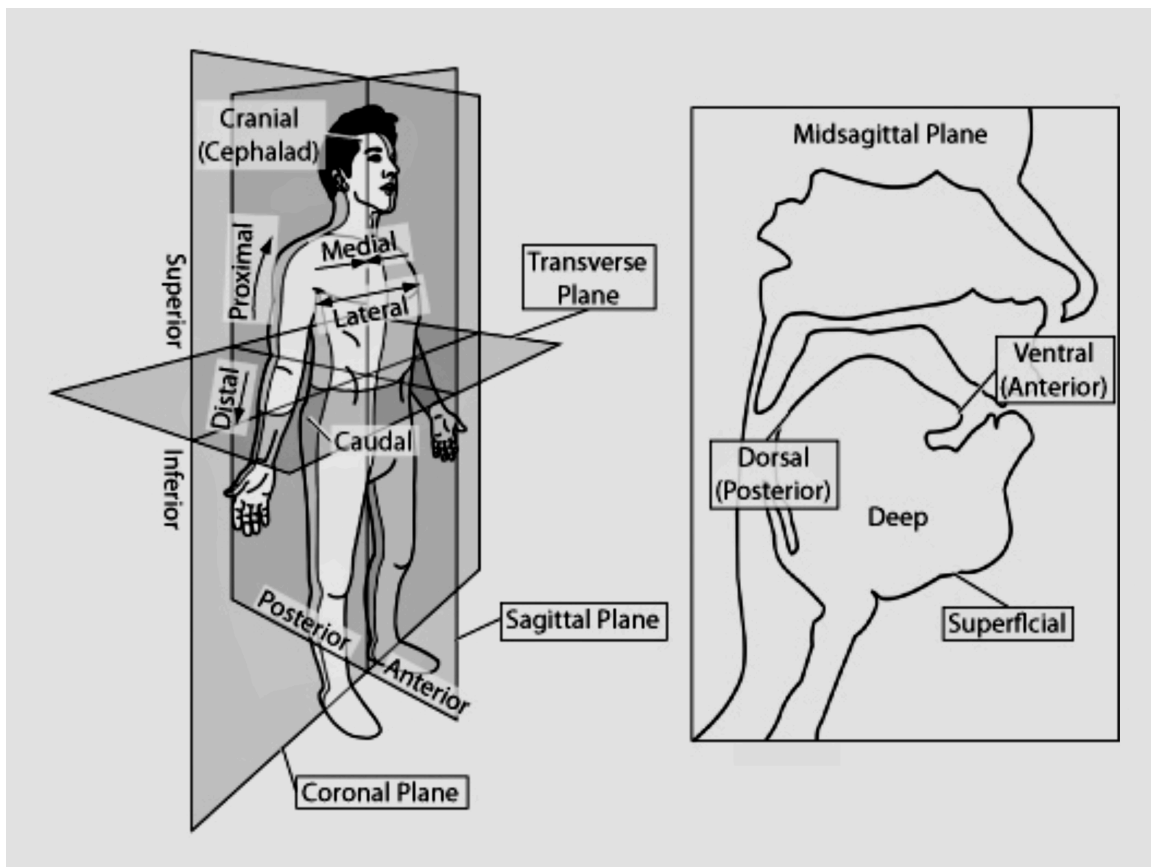
3

Seeing speech

Tools of the trade

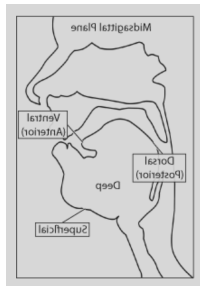
videos/xray.webm

Conceptualizing the head





Conceptualizing the head



CONSONANTS (PULMONIC)

© 2020 IPA

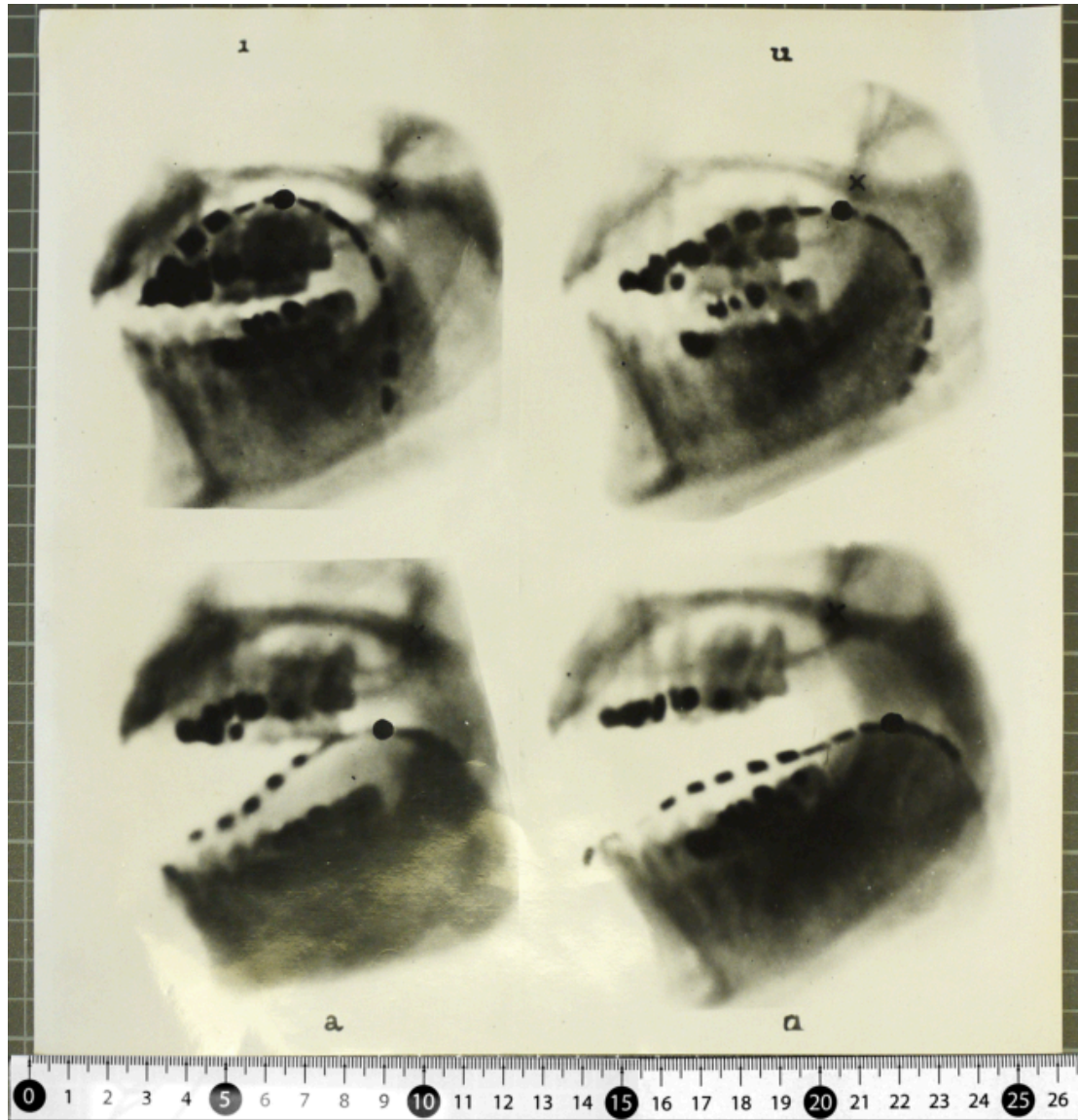
	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.



Tools of the trade

X-Ray





Tools of the trade

Ultrasound tongue imaging

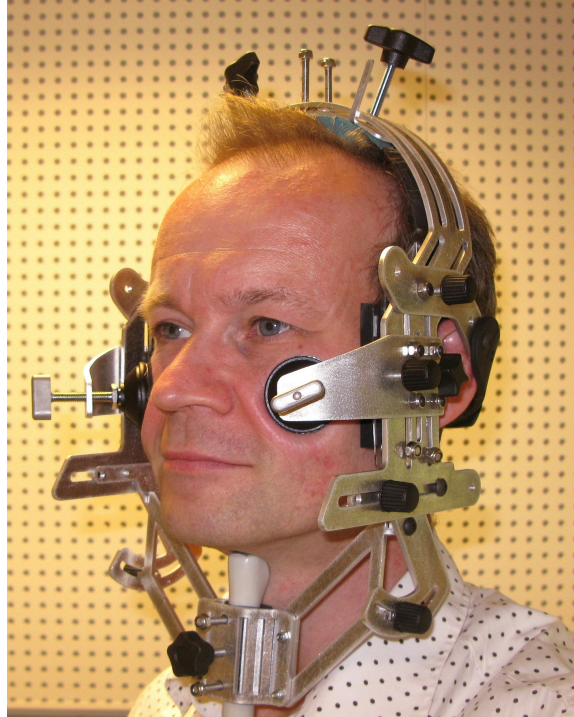


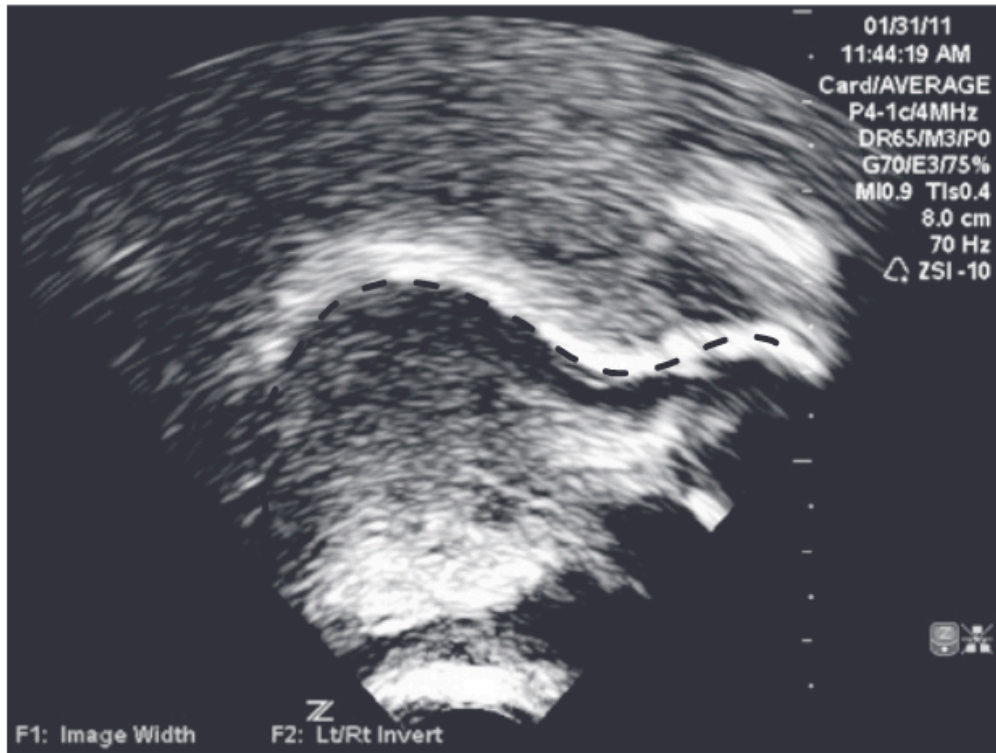
Figure 1: source: <https://www.articulateinstruments.com/ultrasound-imaging/?target=Micro%2FSonoSpeech>

Tools of the trade

Ultrasound tongue imaging

posterior ←

→ anterior



S. Lin, P. S. Beddor, and A. W. Coetzee [1]

Tools of the trade

Electromagnetic Articulography (EMA)

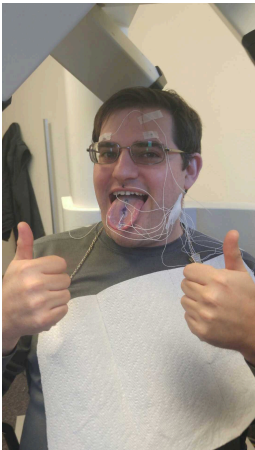
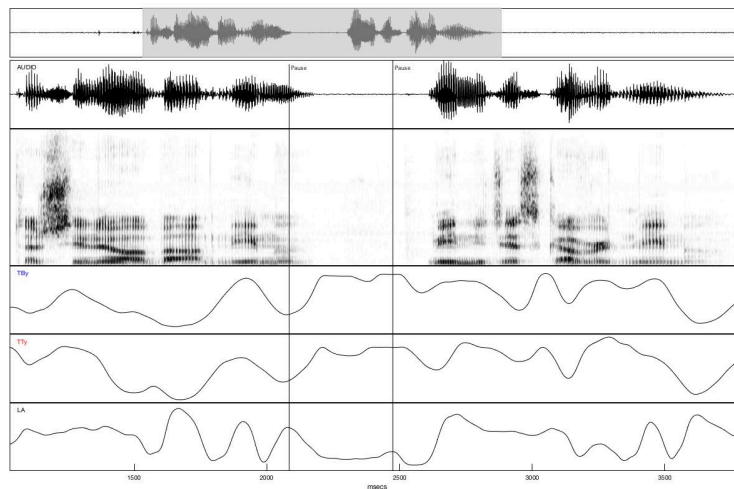


Figure 1: Credit:
Will Styler





Tools of the trade

MRI

videos/mri.mp4

Moving air around

Option 1: The lungs (pulmonic)

- On the out-breath: “egressive”
- On the in-breath: “ingressive”

Moving air around

Ingressive:

~~[...] no language uses the pulmonic ingressive airstream in any systematic way.~~

Many languages, including some Englishes, use ingressive speech for “backchanneling”.

Terminology

- “Sublaryngeal”: below the larynx
- “Supralaryngeal”: above the larynx

Larynx and the voice



videos/mel_blanc.mp4

pos-
terior

anterior



References

Bibliography

- [1] S. Lin, P. S. Beddor, and A. W. Coetzee, “Gestural reduction, lexical frequency, and sound change: A study of post-vocalic /l/,” vol. 5, no. 1, pp. 9–36, 2014, doi: 10.1515/lp-2014-0002.